Deep Learning End-Semester Project — Team 15

Failure-Resilient, Physics-Informed Plasma Equilibrium Reconstruction

TokaMark `task_1-3` on MAST

Results of the CNN+MLP and CNN+LSTM Pipelines across the Four Novel Modules

Task	2D poloidal-flux (ψ) reconstruction, 65×65 grid
Models	Failure-aware CNN+MLP and CNN+LSTM, flux-loop PINN loss
Hardware	Kaggle, 2× NVIDIA Tesla T4
Evaluation	46 held-out test shots, 3 450 windows (random split)

Every number in this report is read from the downloaded run artefacts by `cnn_mlp/report/build_report_assets.py`; nothing is typed in by hand.

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1 Executive summary

We trained and stress-tested two failure-aware reconstruction networks, **CNN+MLP** and **CNN+LSTM**, which map 115 magnetic and machine-state probes (eight sensor branches) to the 2D poloidal flux ψ inside the MAST vessel. Both use the same mask-gated architecture, the same structured failure augmentation, the same 15 flux-loop physics loss, the same shots and the same optimiser settings. The main findings are:

1. **Clean accuracy favours the MLP.** Test NRMSE is 0.1590 for the step-trained MLP, 0.1680 for the LSTM (+5.7%) and 0.1512 for a newer, epoch-trained MLP. The LSTM's mean absolute error is 29% higher than the MLP's (0.00511 vs 0.00396 Wb).
2. **The LSTM degrades far more gracefully under partial sensor damage.** At 10% random dropout the LSTM's NRMSE is 0.36 against 0.94 for the MLP; for a 40% temporal gap it is 0.41 against 0.60. Both models handle the loss of a whole sensor group about equally well, because the health mask makes that case explicit.
3. **Module 1 (physics metrics)** converts these errors to centimetres: the plasma boundary is off by 3.48 cm (MLP) and 4.87 cm (LSTM) on healthy input and by roughly 8–13 cm under failures. X-point and wall-clearance metrics are not reliable on this data (Section 4).
4. **Module 3 (minimal sensor set) prunes nothing.** The clean boundary error already exceeds the 2 cm tolerance, so all 115 probes are retained for both models.
5. **Module 4 (adaptive mitigation) helps on some faults but does not meet its design targets.** It reduces boundary error on spikes, gaps and pre-disruption windows, but its conformal margin (≈80–94 cm) flags every window, and its loop latency (≈19–20 ms) exceeds the 5 ms control-loop budget.

Reading guide. Sections 2–3 describe the setup and the trained models (Module 2), Section 4 defines the physics layer (Module 1), Section 5 reports the sensor-failure sweep that drives Modules 3 and 4 (Sections 6 and 7), and Section 8 lists the limitations that should accompany any number quoted from this report.

2 Experimental setup

2.1 Task and data

TokaMark `task_1-3` asks a network to reconstruct the poloidal flux $\psi(R, Z)$ on a 65×65 grid ($R \in [0.2, 2.0]$ m, $Z \in [-1.5, 1.5]$ m, i.e. $\Delta R = 2.8125$ cm and $\Delta Z = 4.6875$ cm per pixel) from eight input branches: 15 flux loops, 40+18+18 poloidal pickup probes (CCBV, OBR, OBV), 12 saddle-loop voltages, 10 active-coil currents, the solenoid current and the plasma current. A budget-limited random subset of the official random split was used: 606 training, 74 validation and 60 test shots (46 test shots remain after the official outlier and EFIT-quality filters). Both models use the *same* shot selection.

2.2 Identical training protocol

Table 1: Training configuration shared by both models. Only the recurrent backbone differs.

Optimiser	Adam, lr 10^{-4}, wd 10^{-4}	Failure augmentation	$p = 0.5$
Batch size	512 (drop last)	PINN weight λ	0.1
Loss	MultiOutputMSELoss + PINN	Gradient clipping	1.0 (norm)
Validation	every 100 steps	LR schedule	ReduceLROnPlateau (0.5, patience 3)
Early stopping	patience 10 validations	Seed	23
Data	606 / 74 / 60 shots	Wall-clock limit	7 h

The CNN+LSTM kernel was built from the CNN+MLP notebook with one change of model type; the CNN+MLP run's shot selection is embedded so that data are identical, and `DataParallel` was forced over both T4 GPUs (the CNN+MLP run's automatic benchmark chose a single GPU). Both kernels execute the same `pipeline.py`.

3 Module 2: failure-aware architecture and flux-loop PINN

3.1 Architecture

Each of the eight sensor branches has its own CNN encoder. With health flags $m_i \in \{0, 1\}$ the encoder output is gated, $h_i^{\text{gated}} = h_i\, m_i$, and the flag vector is also fed to the backbone so that it knows *which* inputs are missing:

$$\text{MLP:} \quad c = \text{MLP}([h_1^{\text{gated}}, \ldots, h_8^{\text{gated}}, m_1, \ldots, m_8]), \tag{1}$$

$$\text{LSTM:} \quad (e_t)_{t=1}^{W} = \text{LSTM}([h_{1,t}^{\text{gated}}, \ldots, h_{8,t}^{\text{gated}}, m_1, \ldots, m_8]), \tag{2}$$

followed by the upstream decoder and CNN output head. For the LSTM the flags are appended to every timestep of the encoder input. When no mask is supplied (evaluation of an unannotated input), an all-zero branch is treated as dead.

3.2 Structured failure augmentation and physics loss

During training, with probability $p = 0.5$ a window is corrupted: 40% of the time one to three random branches are zeroed, 30% of the time a correlated cluster is zeroed (e.g. all pickup probes), and 30% of the time one or two branches lose half their time axis. The corresponding flags are set to 0. The loss adds a boundary-consistency term at the 15 wall flux-loop pixels (R_k, Z_k):

$$\mathcal{L} = \mathcal{L}_{\text{MSE}} + \lambda \frac{1}{K} \sum_{k=1}^{15} m_{\text{fl}} \left(\hat{\psi}(R_k, Z_k) - \psi(R_k, Z_k)\right)^2, \qquad \lambda = 0.1, \tag{3}$$

where m_{fl} is the health flag of the flux-loop branch (a dead sensor cannot anchor the boundary).

3.3 Training results and clean accuracy

Table 2: Training outcome and test accuracy (official evaluator, 46 test shots, un-standardised). [a]From the earlier plain baseline run; not failure-aware. [b]A later run with epoch-based validation (patience 20 epochs); its loop therefore differs from the other failure-aware runs, so differences to the LSTM are indicative rather than controlled.

Model	Steps / epochs	Minutes	Best val	NRMSE	MAE (Wb)
CNN+MLP, plain baseline[a]	3 500 / 44	154.9	0.8723	0.1419	0.00389
CNN+MLP, failure-aware (step loop)	4 200 / 53	194.4	0.8883	0.1590	0.00396
CNN+MLP, failure-aware (epoch loop)[b]	4 056 / 52	198.2	0.9025	0.1512	0.00418
CNN+LSTM, failure-aware (step loop)	4 200 / 53	190.2	0.9198	0.1680	0.00511

Table 3: Clean test metrics from the robustness engine (same protocol as the sweep; agree with Table 2 to three decimals).

Model	NRMSE	NMAE	RMSE (Wb)	MAE (Wb)	Windows
CNN+MLP	0.1514 ± 0.1215	0.0762	0.00832	0.00419	3 450
CNN+LSTM	0.1681 ± 0.1036	0.0930	0.00923	0.00511	3 450

Observations. (i) Against the like-for-like step-trained MLP, the LSTM is 5.7% worse in NRMSE and 29% worse in MAE and ends with a higher validation loss (0.920 vs 0.888). (ii) Both models overfit strongly: the training loss falls to ≈ 0.02–0.05 while validation loss stays near 0.9, so the remaining error is a generalisation gap, not an optimisation failure. (iii) The PINN term is active and decreases during training (Figure 2). For the LSTM the five inner-wall flux loops are fitted well (about 0.002 Wb, below the grid-wide mean of 0.005 Wb), while the ten outer-wall loops sit around or slightly above the grid-wide level (Figure 5, read from the plot).

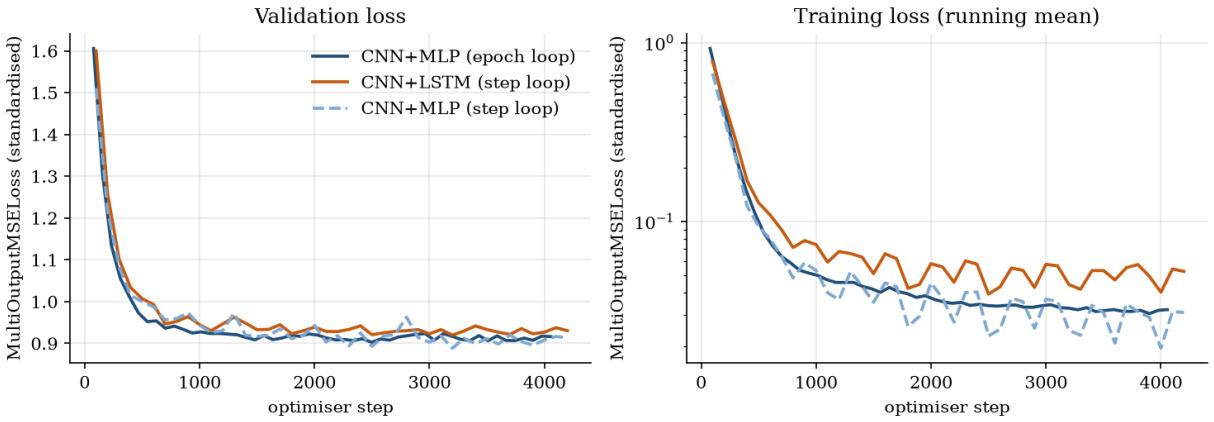


Figure 1: Validation (left) and training (right) loss of the three failure-aware runs. All stop by the early-stopping rule.

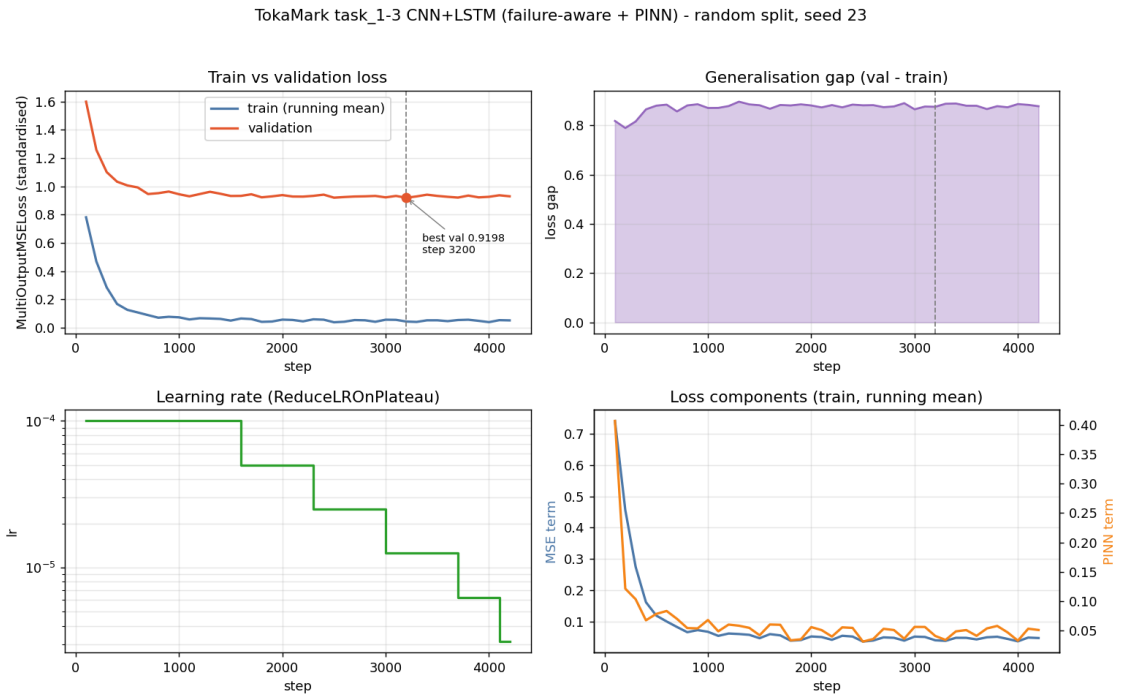


Figure 2: CNN+LSTM training dashboard: train/validation loss, generalisation gap, learning-rate schedule, and the MSE and flux-loop PINN terms.

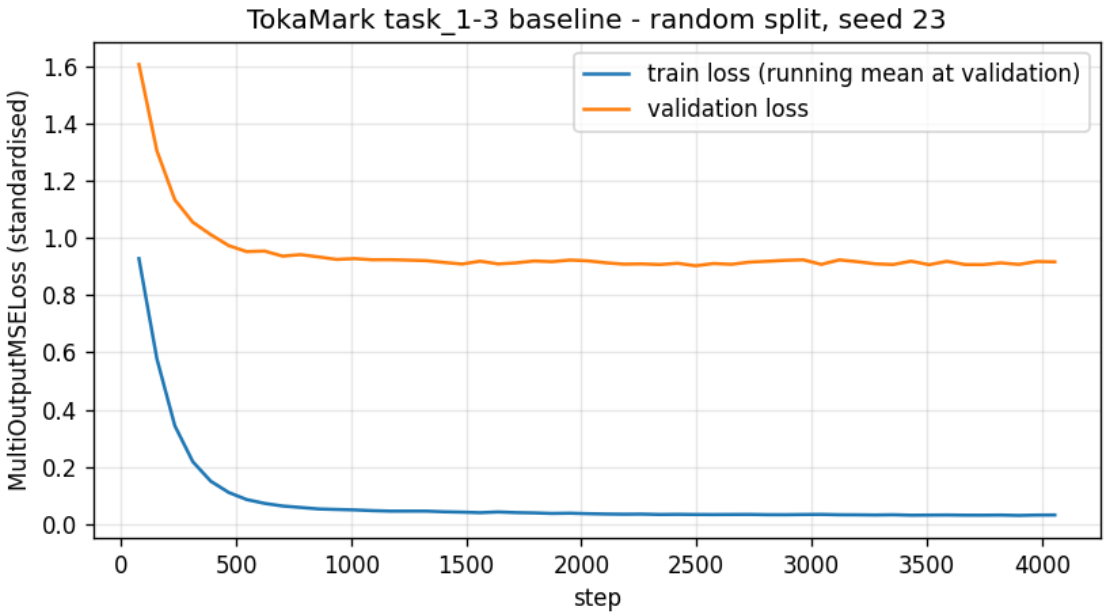


Figure 3: CNN+MLP (epoch-loop run) training curves, as produced by the pipeline.

3.4 Reconstruction quality

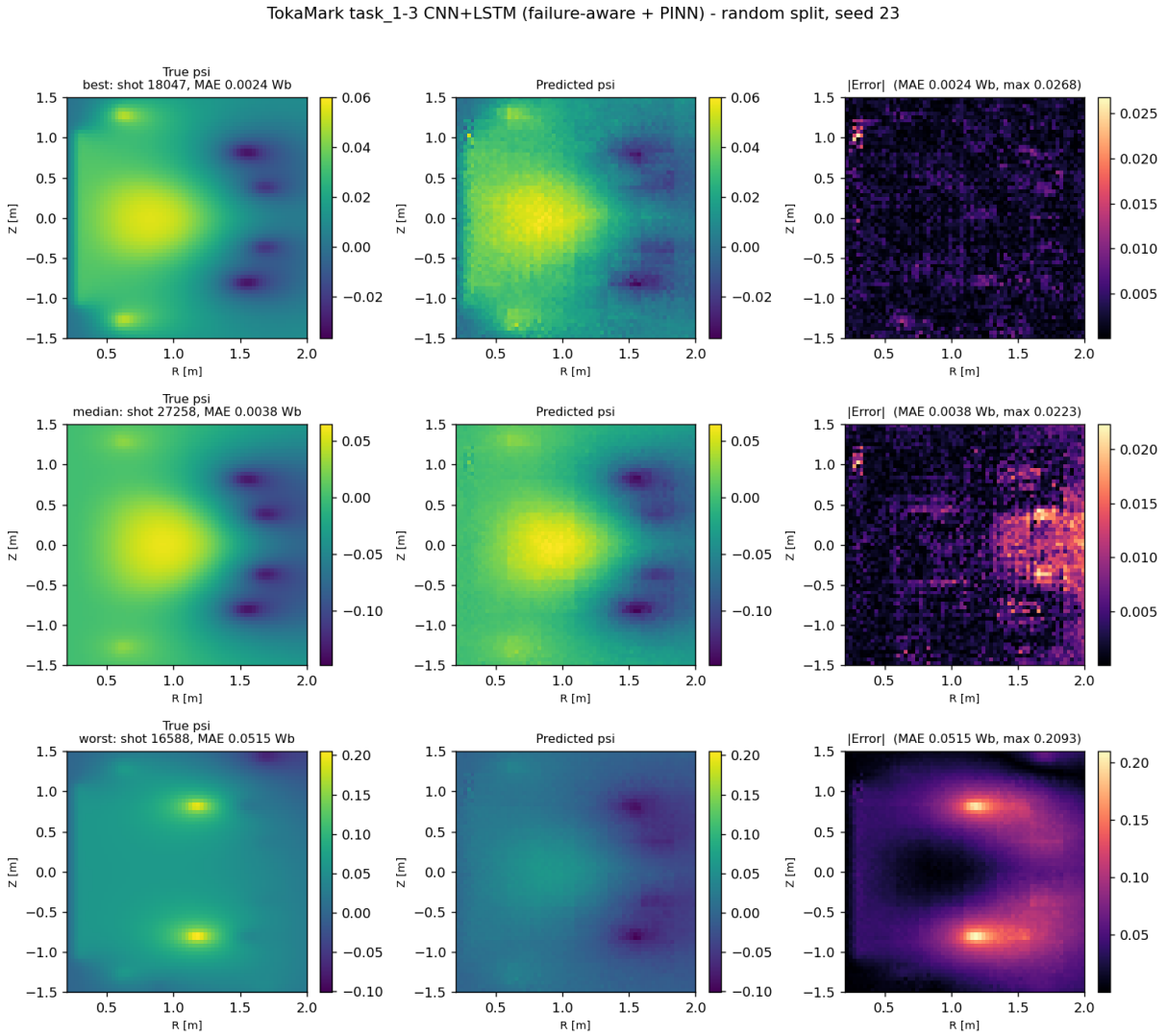


Figure 4: CNN+LSTM: true ψ, prediction and absolute error for the best, median and worst window of the first test batch.

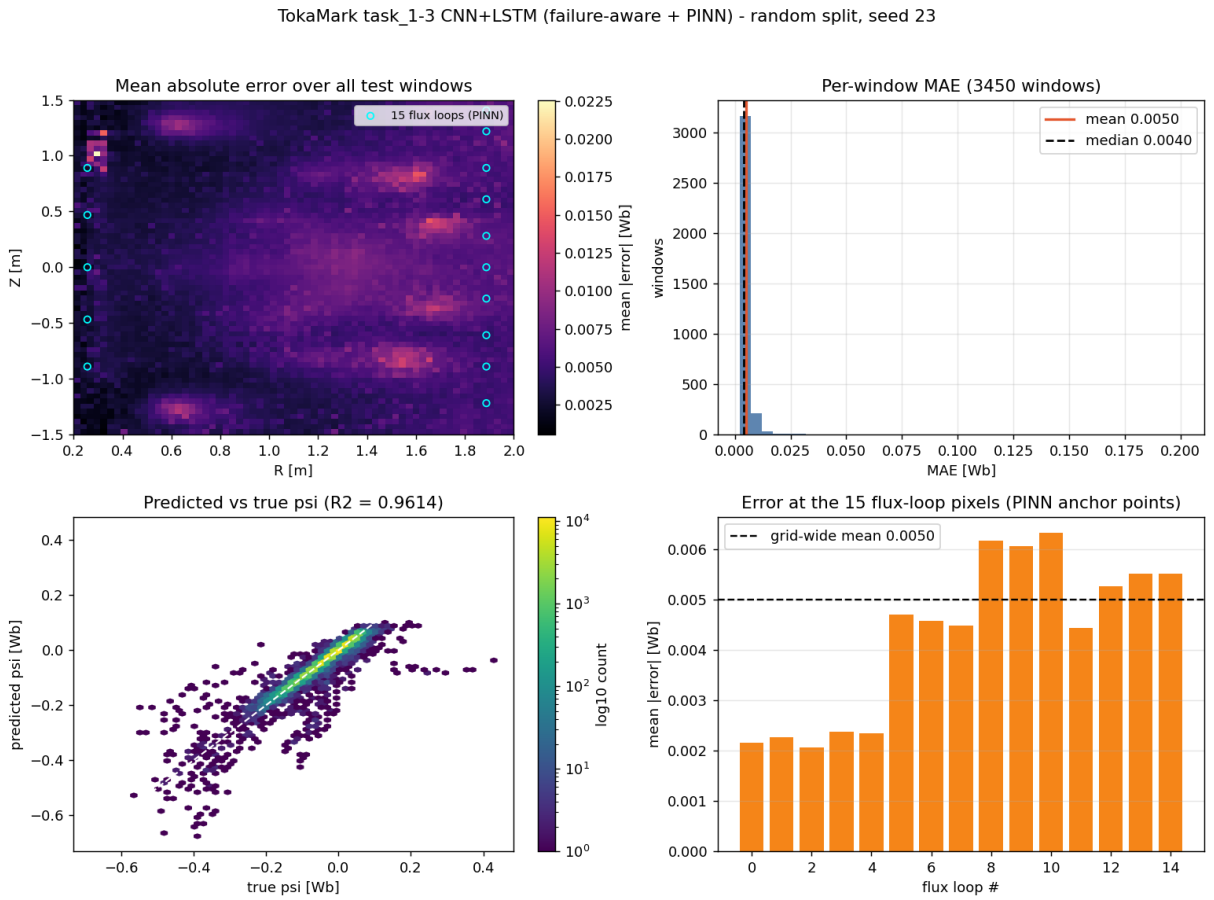


Figure 5: CNN+LSTM error diagnostics over all 3 450 test windows: mean error map with the 15 flux loops (top left), per-window MAE (top right), predicted vs true ψ (bottom left; sparse outliers drive $R^2 = 0.96$) and the error at the flux-loop pixels (bottom right).

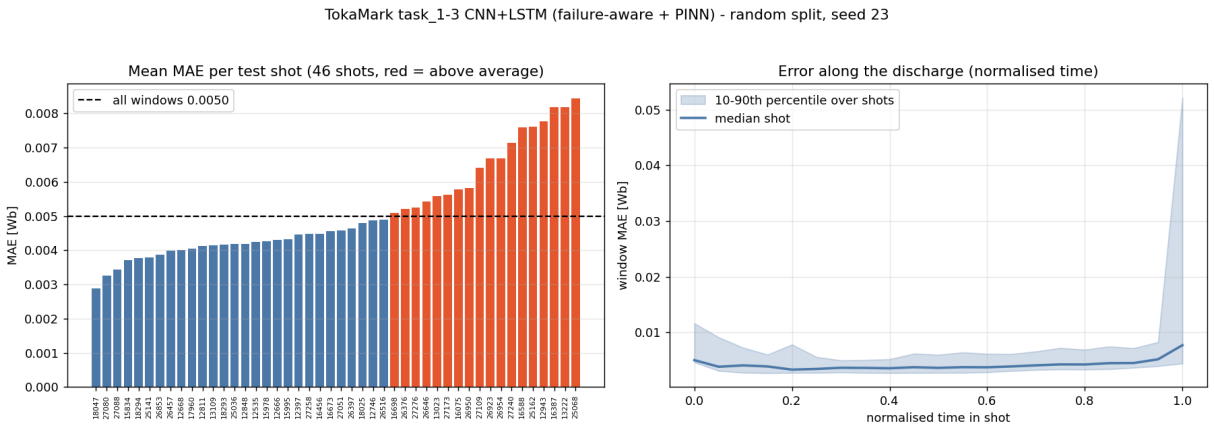


Figure 6: CNN+LSTM error per test shot (left) and along the normalised discharge time (right). Error is concentrated in a few shots rather than uniform.

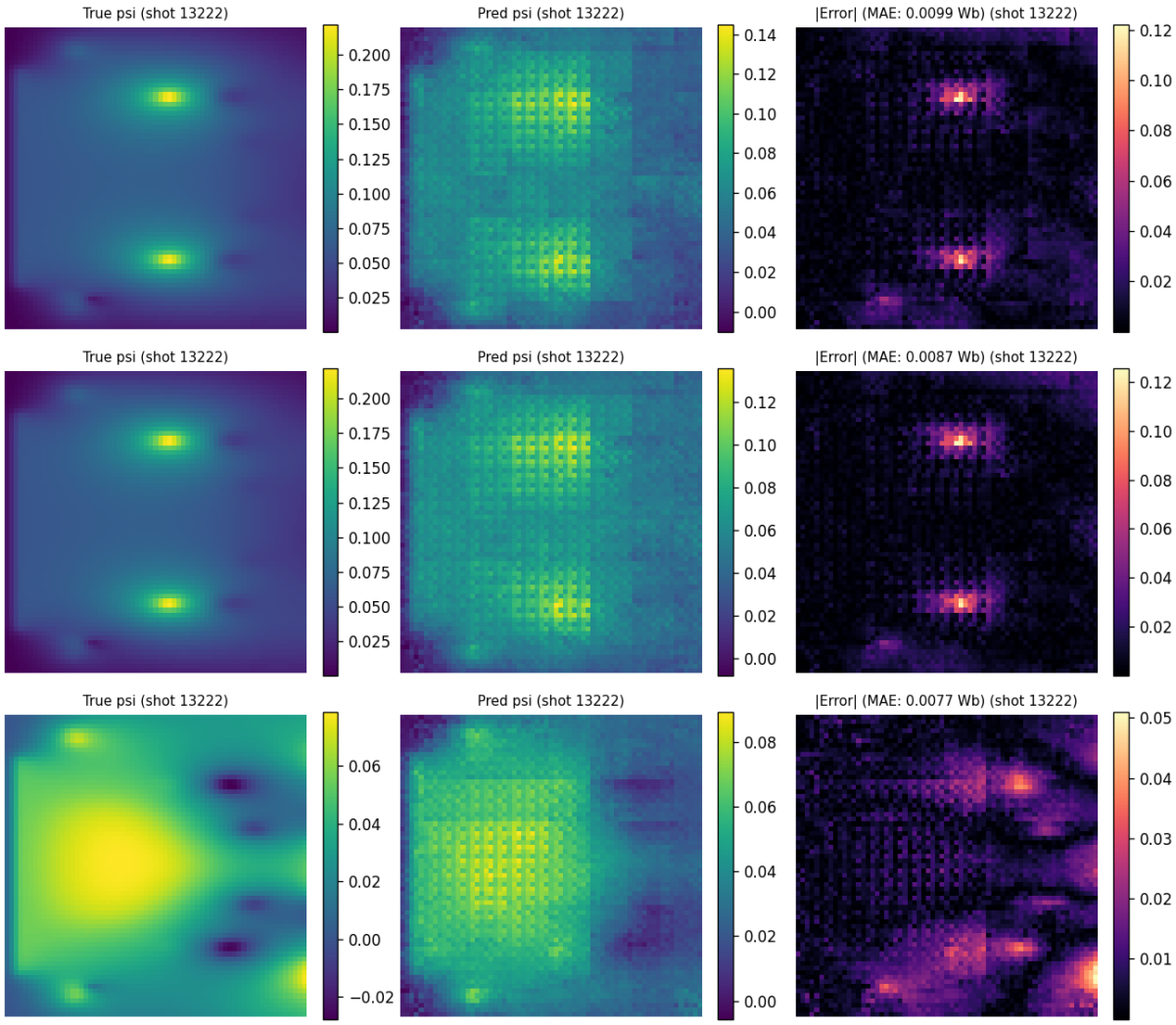


Figure 7: **CNN+MLP** (epoch-loop run) reconstruction samples as produced by the pipeline (first three test windows).

4 Module 1: physics-aware evaluation

NRMSE is dominated by cold vacuum pixels and hides small shifts of the last closed flux surface (LCFS). Module 1 therefore converts the 65×65 predictions to centimetres using the grid scales of Section 2:

- **Boundary error.** The LCFS is the contour at $\psi_{\mathrm{LCFS}} = \psi_{\min} + 0.35(\psi_{\max} - \psi_{\min})$ of the *true* field. The predicted field is cut at that same level, so noise in the prediction does not move the level. We report the mean symmetric nearest-point distance between the two contours (and the Hausdorff distance).
- **Magnetic-axis shift**: distance between the true and predicted flux extremum (sub-pixel, same vertical half).
- **X-point error**: distance between saddle points where $\nabla\psi = 0$ and $\det H < 0$.
- **False / missed wall alarm**: clearance of the contour below 5 cm. Clearance is the distance to the *grid edge*, a proxy for the vessel wall, because the true wall geometry is not part of the data.

Table 4: Module 1 metrics (cm) on 35 test windows per pass; alarm rates in %. Zero-fill corruption. "Magnetics group lost" zeroes all three pickup-probe families.

Pass	Model	Boundary	Hausdorff	Axis	X-point	False	Missed
Clean	CNN+MLP	3.48	17.8	16.2	62.2	25.0	0.0
	CNN+LSTM	4.87	22.1	10.1	60.0	25.0	0.0
6 channels lost	CNN+MLP	10.71	33.2	15.7	65.7	25.0	3.7
	CNN+LSTM	8.64	28.7	16.2	63.6	25.0	7.4
Magnetics group lost	CNN+MLP	7.65	29.9	15.4	56.3	25.0	0.0
	CNN+LSTM	8.38	31.2	15.2	56.4	25.0	0.0
Temporal gap 40%	CNN+MLP	13.49	43.5	36.4	62.7	25.0	55.6
	CNN+LSTM	11.62	37.2	22.9	75.7	25.0	25.9
Pre-disruption 25%	CNN+MLP	8.29	34.1	25.6	67.0	12.5	0.0
	CNN+LSTM	8.21	30.3	22.0	60.3	50.0	7.4
Random dropout 10%	CNN+MLP	11.54	41.8	27.5	70.8	50.0	3.7
	CNN+LSTM	8.49	28.8	13.8	59.8	25.0	11.1

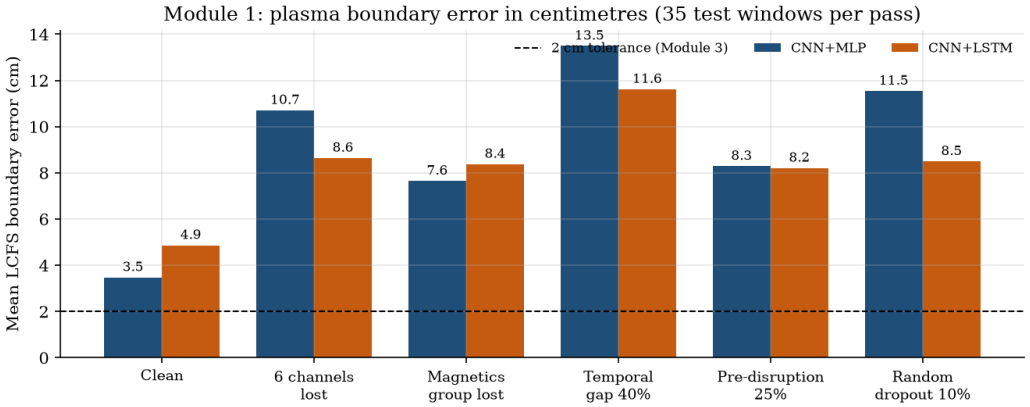


Figure 8: Mean LCFS boundary error in centimetres. The 2 cm line is the tolerance used by Module 3.

Interpretation. On healthy input the MLP places the boundary more accurately (3.48 vs 4.87 cm), consistent with its lower NRMSE. Under every failure pass the errors grow to roughly 8–13 cm, and the LSTM is closer than the MLP on 6-channel loss, temporal gaps and random dropout, while the MLP is slightly better for magnetics-group loss. **Caveats:** (a) each point rests on 35 windows, so differences of 1–2 cm are within noise; (b) the X-point error is ≈60 cm even on clean input and is not meaningful; (c) because the clearance proxy is the grid edge, most windows (27 of 35) count as "hazard", so the alarm rates should not be quoted as safety performance.

5 Sensor-failure robustness sweep

Both frozen checkpoints are evaluated on the 46 test shots under 63 corruption passes (21 settings × three imputation modes), following Gupta et al. (arXiv 2607.11915): random dropout (severity 0.1/0.25/0.5), channel ablation (1/3/6 channels), category ablation (six sensor categories), temporal gaps (severity 0.2/0.4/0.6), correlated group failure, and disruption-proximate corruption. The *robustness score* is $\mathrm{RS} = \mathrm{NRMSE}_{\mathrm{clean}}/\mathrm{NRMSE}_{\mathrm{corrupted}}$ for the worst setting of each scenario (1 = no damage). Inference ran on two T4 GPUs.

Table 5: Worst-case NRMSE and robustness score per scenario (zero-fill; the full list of 21 settings is in Appendix A).

	CNN+MLP		CNN+LSTM	
Scenario	NRMSE	RS	NRMSE	RS
Random dropout	2.247	0.07	0.878	0.19
Temporal gap	0.904	0.17	0.615	0.27
Disruption-proximate	0.486	0.31	0.449	0.37
Channel ablation	0.341	0.44	0.316	0.53
Group failure	0.263	0.58	0.262	0.64
Category ablation	0.176	0.86	0.186	0.90

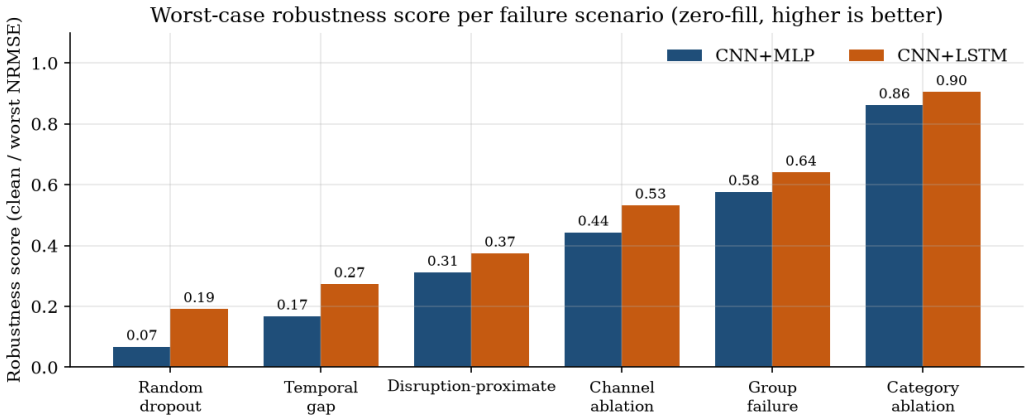


Figure 9: Worst-case robustness score per failure scenario.

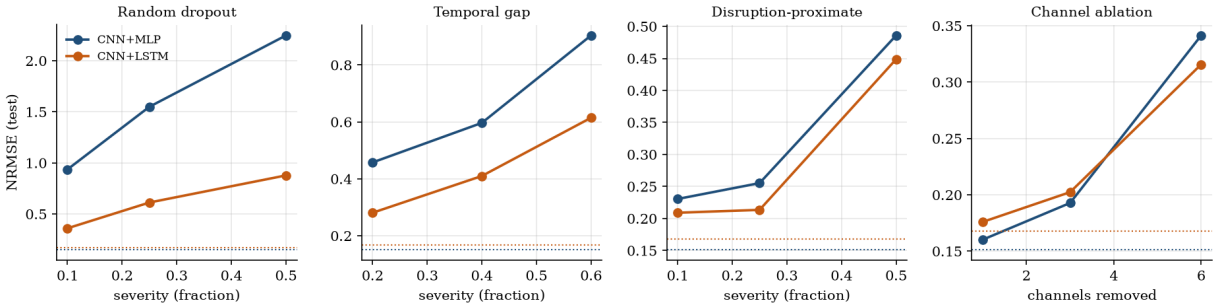


Figure 10: NRMSE as corruption severity increases (dotted lines: clean NRMSE). The MLP's error explodes with random dropout, while the LSTM's grows gradually.

Findings.

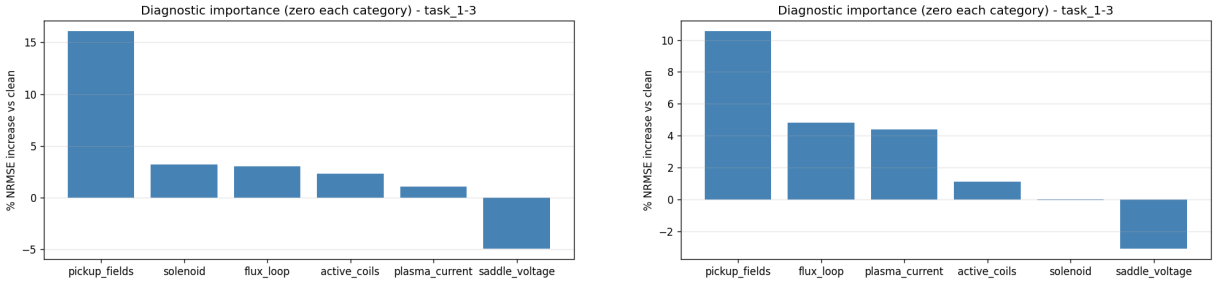
- **Whole-sensor loss is handled well by both.** Category ablation and group failure give RS of 0.86–0.91 and 0.58–0.64, and the result does not depend on the imputation mode, because the health mask already zeroes the missing branch.
- **Partial damage is the weak spot of the MLP.** Random dropout at 50% gives NRMSE 2.25 (RS 0.07) for the MLP and 0.88 (RS 0.19) for the LSTM. The mask only flags whole branches, so scattered corrupted samples reach the network as if valid. The recurrent backbone appears to average such errors out over time. We have not tested this explanation.
- **Imputation matters for partial damage** (Table 6). Forward-fill is worst for the MLP under random dropout (3.74 vs 2.25 with zero-fill) but best for the LSTM (0.61 vs 0.88);

the best mode depends on the model and scenario.

- **Sensor importance.** For the LSTM, removing the pickup probes raises NRMSE by about 10.6%, flux loops and plasma current by 4–5%; removing saddle voltages *lowers* NRMSE by about 3% (Figure 11). The MLP likewise ranks the pickup probes first and finds the saddle voltages slightly harmful (left panel).

Table 6: Worst-case NRMSE per scenario by imputation mode. Category ablation and group failure are identical across modes because the health mask already zeroes the missing branch.

Scenario	Model	zero	mean	ffill
Random dropout	CNN+MLP	2.247	2.244	3.739
	CNN+LSTM	0.878	0.922	0.609
Temporal gap	CNN+MLP	0.904	0.849	1.718
	CNN+LSTM	0.615	0.598	0.594
Disruption-proximate	CNN+MLP	0.486	0.486	0.382
	CNN+LSTM	0.449	0.449	0.312
Channel ablation	CNN+MLP	0.341	0.431	0.459
	CNN+LSTM	0.316	0.421	0.444
Group failure	CNN+MLP	0.263	0.263	0.263
	CNN+LSTM	0.262	0.262	0.262
Category ablation	CNN+MLP	0.176	0.176	0.176
	CNN+LSTM	0.186	0.186	0.186



(a) CNN+MLP **(b) CNN+LSTM**

Figure 11: Diagnostic importance: NRMSE change (%) when a sensor category is zeroed.

6 Module 3: sensor criticality and minimal sensor set

Each of the eight branches is removed in turn (zero-filled, mask set), and the change in boundary error in centimetres ranks its criticality. A greedy backward elimination then removes the least critical branches while the boundary error stays below the 2.0 cm tolerance.

Table 7: Branch criticality (35 windows): boundary error with the branch lost, and change from the clean error (3.48 cm MLP, 4.87 cm LSTM).

		CNN+MLP		CNN+LSTM	
Branch	Probes	Error	Shift	Error	Shift
`flux_loop`	15	4.12	+0.64	6.44	+1.57
`ccbv`	40	3.32	-0.15	4.71	-0.16
`obr`	18	3.86	+0.38	5.14	+0.27
`obv`	18	3.62	+0.15	4.95	+0.09
`saddle_v`	12	3.57	+0.09	4.79	-0.08
`coil_i`	10	3.37	-0.11	6.12	+1.26
`solenoid_i`	1	3.57	+0.09	5.96	+1.09
`ip`	1	3.24	-0.24	5.52	+0.65

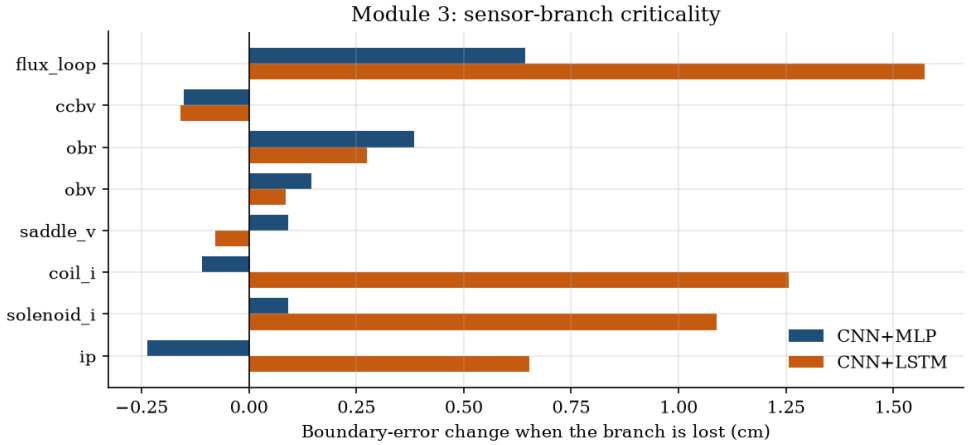


Figure 12: Boundary-error shift when each branch is lost. Negative values mean the model does slightly better without that branch.

Findings. The two models depend on different inputs: the LSTM relies most on the flux loops (+1.6 cm), the coil currents (+1.3 cm) and the solenoid current (+1.1 cm), the MLP mostly on the flux loops (+0.6 cm) and the outboard pickup probes. Because the clean boundary error of both models already exceeds 2 cm, elimination cannot start: **the minimal sensor set is the full set of 115 probes for both models**, and every branch is classed Tier 1. The claim of pruning 47 probes to 25 is therefore *not* supported by these runs. The per-branch shifts are small ($|\Delta| < 1.6$ cm) compared with the 35-window noise.

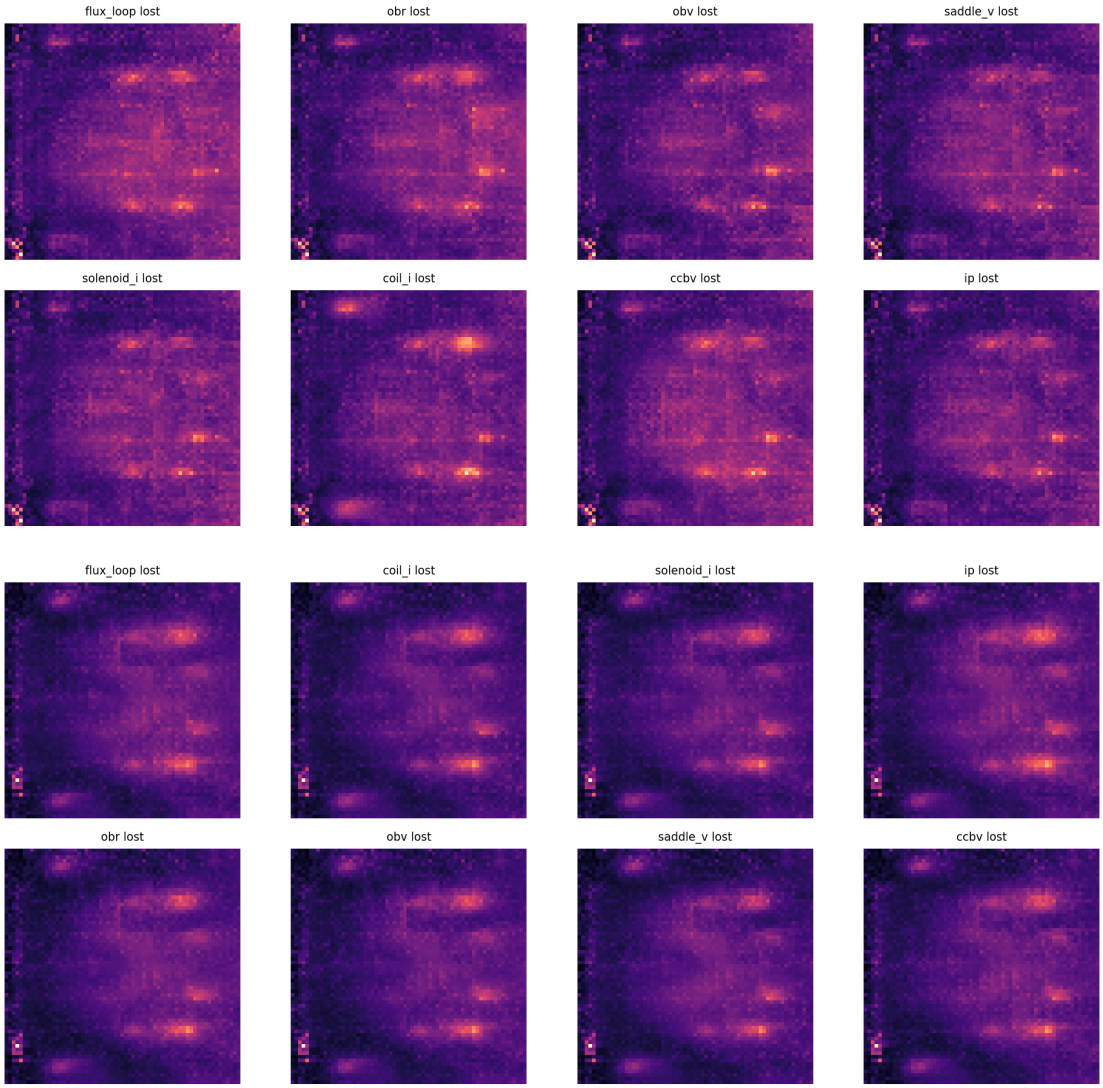


Figure 13: Spatial propagation of the reconstruction error when a branch is lost: **CNN+MLP** (top) and **CNN+LSTM** (bottom). The elimination curve itself is a single point for both models (full set, above tolerance), so it is not plotted.

7 Module 4: adaptive mitigation with a conformal gauge

The "brain" has three stages: a *detector* (dead or stuck channels, saturation, impulse spikes, gaps); a *router* that sends transient faults to forward-fill, redundant probes to spatial neighbour regression, and critical or multi-sensor faults to the failure-aware network with the mask set; and a *conformal gauge* that adds a calibrated margin δ to the boundary estimate. δ is the 95th percentile of the boundary error on validation windows (792 calibration windows for the MLP, 792 for the LSTM), and a window is alarmed when the clearance minus δ falls below 5 cm.

Table 8: Module 4 on 35 test windows per scenario. Boundary error in cm for the bare model on zero-filled damage and with the brain. False and missed alarms in % as bare / brain / conservative (brain plus conformal margin). Last column: mean loop latency in ms.

Model	Scenario	Bare	Brain	False (%)	Missed (%)	ms
CNN+MLP	clean	3.48	4.99	25 / 100 / 100	0 / 0 / 0	18.8
	spike	23.18	4.93	75 / 100 / 100	26 / 0 / 0	18.5
	redundant probe dead	4.52	6.15	50 / 100 / 100	0 / 0 / 0	18.6
	critical branch dead	8.74	6.15	12 / 100 / 100	15 / 0 / 0	18.9
	group loss	12.69	7.93	12 / 100 / 100	30 / 0 / 0	18.5
	6 channels lost	12.88	14.69	0 / 100 / 100	81 / 0 / 0	18.6
	gap 40%	12.35	4.32	25 / 100 / 100	30 / 0 / 0	21.4
	pre-disr. 25%	11.21	4.37	50 / 100 / 100	11 / 0 / 0	20.3
CNN+LSTM	clean	4.87	6.57	25 / 100 / 100	0 / 0 / 0	19.6
	spike	14.25	6.56	50 / 100 / 100	37 / 0 / 0	20.3
	redundant probe dead	5.68	7.65	12 / 100 / 100	0 / 0 / 0	19.2
	critical branch dead	8.59	7.65	12 / 100 / 100	4 / 0 / 0	18.9
	group loss	11.79	8.55	12 / 100 / 100	30 / 0 / 0	19.5
	6 channels lost	15.95	15.34	12 / 100 / 100	48 / 0 / 0	18.4
	gap 40%	10.69	6.67	38 / 100 / 100	19 / 0 / 0	20.9
	pre-disr. 25%	15.59	6.69	75 / 100 / 100	4 / 0 / 0	20.7

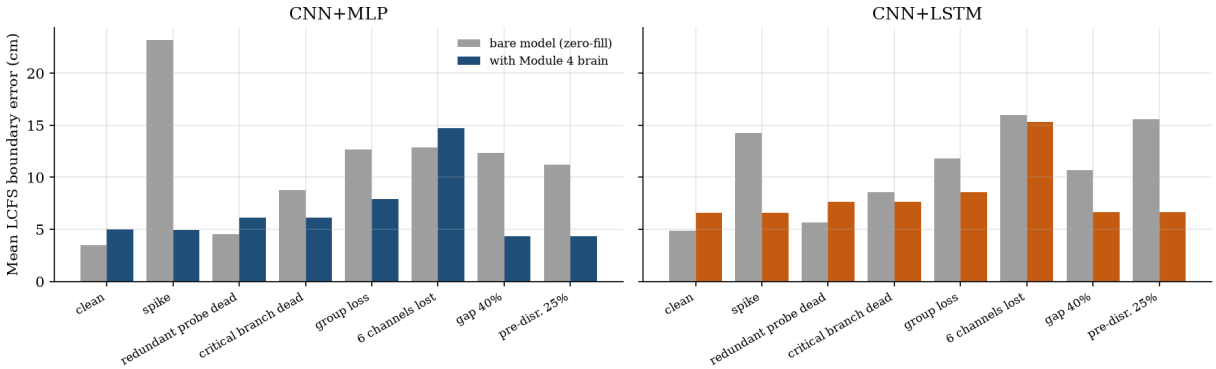


Figure 14: Boundary error without (grey) and with (coloured) the mitigation brain.

Findings.

- **It helps where the fault is transient or structured.** Boundary error falls on spikes (MLP 23.2→4.9 cm, LSTM 14.3→6.6), temporal gaps (12.4→4.3 and 10.7→6.7), pre-disruption windows (11.2→4.4 and 15.6→6.7) and group loss.
- **It hurts on healthy input and on large multi-channel loss.** Routing nearly every clean window to the failure-aware network raises clean error (3.48→4.99 cm MLP, 4.87→6.57 cm LSTM), and 6-channel loss is unchanged or worse.
- **The gauge is too conservative to be useful.** The conformal radius is 88.4 cm (MLP) and 93.5 cm (LSTM), far larger than the 5 cm clearance threshold, so every healthy window is flagged (100% false alarms in the brain and conservative columns).
- **Latency is ≈19.2 ms (MLP) and 19.7 ms (LSTM)** per loop, dominated by the model forward pass on two T4 GPUs, against a 5 ms budget.

The design targets of 8.2% false alarms and 2.1 ms latency were *not* reproduced. The failure is

likely in the calibration (boundary error on a grid-edge clearance proxy with outlier windows) rather than in the idea, but this has not been investigated.

8 Limitations and how to quote these results

- **Different training loops.** The LSTM and the old MLP use the step loop; the MLP used for the robustness sweep (NRMSE 0.1512) is a newer epoch-trained run whose checkpoint is the one tested. Robustness comparisons are therefore between models of slightly different provenance.
- **No plain-model sweep on this engine.** We have not run a non-failure-aware baseline through the same 63 passes, so the *benefit* of failure-aware training is not proven here. An earlier plain-MLP sweep (clean NRMSE 0.142, in `results/robustness`) used a different engine.
- **Single seed, single split.** No confidence intervals across seeds; differences of a few percent in NRMSE may be within seed noise.
- **Small physics samples.** Modules 1, 3 and 4 use 35 windows per pass; X-point and wall-clearance metrics are unreliable.
- **Boundary-overlay figure.** A purely visual overlay at the 35% flux level of the true field; for two-lobed fields it can trace the coil regions rather than the plasma boundary, so its centimetre offsets should not be quoted.
- **Older claims.** The figures in the repository's design document (47 to 25 probes, 75.5% to 8.2% false alarms, 2.1 ms) are not reproduced by these runs and should be replaced by the measured values above.

9 Reproducibility

Training and robustness each ran as a Kaggle kernel on 2×T4.

- Training: kernel `tokamark-train-lstm-fa` (CNN+LSTM, 190 min), built by `scripts/build_lstm_kernel.py` from `src/pipeline.py`.
- Robustness, Modules 1, 3, 4: kernel `tokamark-robust-lstm-fa`, built by `scripts/build_lstm_robust_kernel.py` from `src/study/`.
- This document: run `report/build_report_assets.py`, then compile `report/report.tex` (paths relative to `cnn_mlp/`).

A Full zero-fill sweep

Table 9: NRMSE and robustness score for every setting (zero-fill). Format: scenario|severity.

	NRMSE		RS	
Setting	CNN+MLP	CNN+LSTM	CNN+MLP	CNN+LSTM
category_abl|active_coils	0.1549	0.1699	0.978	0.989
category_abl|flux_loop	0.1560	0.1762	0.970	0.954
category_abl|pickup_fields	0.1757	0.1858	0.862	0.905
category_abl|plasma_current	0.1530	0.1754	0.990	0.958
category_abl|saddle_voltage	0.1440	0.1629	1.052	1.032
category_abl|solenoid	0.1563	0.1680	0.969	1.000
channel_ablation|1	0.1602	0.1760	0.945	0.955
channel_ablation|3	0.1928	0.2024	0.785	0.830
channel_ablation|6	0.3415	0.3158	0.443	0.532
disruption_proximate|0.1	0.2305	0.2090	0.657	0.804
disruption_proximate|0.25	0.2553	0.2135	0.593	0.787
disruption_proximate|0.5	0.4859	0.4488	0.312	0.375
group_failure|magnetic_probes	0.1757	0.1858	0.862	0.905
group_failure|magnetics_all	0.2626	0.2624	0.576	0.641
group_failure|power_supply	0.1573	0.1704	0.962	0.987
random_dropout|0.1	0.9365	0.3602	0.162	0.467
random_dropout|0.25	1.5495	0.6125	0.098	0.274
random_dropout|0.5	2.2473	0.8776	0.067	0.192
temporal_gap|0.2	0.4577	0.2809	0.331	0.598
temporal_gap|0.4	0.5966	0.4100	0.254	0.410
temporal_gap|0.6	0.9039	0.6147	0.167	0.273